(Please write your Enrollment Number) Enrollment No. 04801172022

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: <B. Tech.>, Semester:<VI>
(May, 2025)

Subject Code: BAI 302	Subject: Natural Language Processing
Time :3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(2.5*8=20)	CO Mappin
	a) What is the purpose of stop word removal in text preprocessing?		CO1
	b) Explain the concept of TF-IDF and its significance in text analysis.		CO1
	c) Define morphology and state its role in Natural Language Processing.		CO2
	d) What is ambiguity in NLP? Briefly explain with an example.		CO2
	e) What is Neural Machine Translation (NMT) and state two different from phrase-based MT?		CO3
	f) What is the purpose of using LSTM in sequential data processing?		CO3
	g) Mention two differences between CNN and RNN in the context of NLP.		CO4
	h) Write the formula for calculating Inverse Document frequency (IDF). Why it is called so?		CO4
	UNIT I		CO Mappin
Q2	Explain the components of Natural Language Processing with suitable examples. How do these components work together in a typical NLP system?	(10)	CO1
Q3	What is parsing in NLP? Describe different parsing techniques and the challenges involved in syntactic analysis.	(10)	CO1
	UNIT II		CO Mappin
Q4	Compare and contrast word2vec, doc2vec, and GLoVe word embedding techniques. Discuss their working principles and applications in NLP tasks.	(10)	CO2
Q5	Explain the different formats used for representing a text corpus. How do these formats influence text preprocessing and model training?	(10)	CO2
	UNIT III		CO Mappin
Q6	Describe various approaches to Machine Translation. Compare rule-based, statistical, and neural methods with examples.	(10)	CO3
Q7	Explain the architecture of the Anusaraka system. How does it handle translation using the interlingua approach?	(10)	CO3
	UNIT IV		CO Mappin
Q8	Explain the working of LSTM and GRU models. Highlight how they overcome the limitations of standard RNNs in language modeling.	(10)	CO4
Q9	Discuss the use of the Natural Language Toolkit (NLTK) for implementing deep learning-based NLP tasks with examples.	(10)	CO4